

Spillover Effects of Grants: Schools in Rural Pakistan

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I. Introduction

Across rural Pakistan, low-cost private schools operate on the edge of financial survival. They face intense competition, volatile revenue streams, and constant exposure to economic and policy shocks that can push them toward closure. In response, external interventions in the form of grants, loans, and subsidies are introduced to strengthen the sector. Yet these interventions often generate spillover effects that reshape an already fragile environment. A grant may lower costs or improve quality for one school, but it can also leave untreated neighbours struggling to keep up. Despite this, there remains limited evidence on how these interventions weave into the fabric of local market structure, ultimately determining which schools survive. Thereby, it becomes imperative to understand whether such policies stabilise schooling markets or unintentionally trigger displacement.

By tracing how grants ripple through local schooling markets and reshape the competitive pressures schools face, this paper looks into the broader system, leading to the research question:

How do unconditional grants to private schools affect the likelihood of exit among untreated schools operating in the same villages?

The paper's hypothesis follows from models of competitive spillovers: grants that strengthen schools may heighten pressure on nearby untreated competitors, increasing their vulnerability, though market-wide improvements could offset these pressures.

To address this, the paper uses panel data from the LEAPS study and employs a difference-in-differences design. This approach compares untreated schools to similar schools before and after the grant intervention, allowing it to isolate changes in survival arising from shifts in local competition. The regressions incorporate baseline school characteristics - including infrastructure, enrollment, age - and measures of village-level market concentration, ensuring that estimated effects reflect the impact of the grant rather than underlying heterogeneity in school strength or competitive context.

The results show that school closures rose sharply in the post-grant period, with exit rates increasing by 9–17 percentage points across specifications. However, untreated schools in treated villages did not become more likely to close than those in control areas. Market structure, as measured by the HHI, did not meaningfully shape these effects. Instead, the main heterogeneity appears at the school level: institutions with stronger baseline infrastructure were more resilient, while older untreated schools were more likely to exit after the grant cycle.

This paper contributes to the literature by combining theoretical insights on competitive spillovers with evidence from a large-scale policy experiment on private schooling in low-income contexts. It shifts the lens by focusing on schools left outside the intervention with varying market structures and provides insights aimed to help design policies that limit exposure to unintended risks for untreated schools.

II. Literature Review

Low-cost private schools play a major role in educating children in many low- and middle-income countries, and Pakistan is no exception. Enrolment in private primary schools has climbed up from under 5 percent to over 30 percent in the decades between 1990 and 2019 (PIHS 1991; World Bank Databank 2021; Akmal, Hares & Perakis 2022). Much of this growth has taken place in rural parts of Pakistan, where public schools are often understaffed, inaccessible, or exist as ‘ghost’ schools (Andrabi, Das & Khwaja 2008). Yet, this expansion is often met with volatility. Over 10-30 percent of private schools in Pakistan and Chile closed over a decade (Andrabi et al. 2025; Grau, Hojman & Mizala 2018), showcasing the extent to which the thin margins these schools operate on - often pushing them towards exit.

A large body of work helps explain this fragility. Schools often operate with minimal infrastructure and remain at the mercy of their fee revenue almost entirely. This leads to existential threats when faced with volatility on both the demand and supply side - for instance a drop in parental income or high turnover rates of teachers (Andrabi et al. 2009; McKenzie & Paffhausen 2019). Therefore, for survival, it becomes imperative to cover fixed costs. When confronted with closures, they have unintended consequences - or spillover effects - across the market. Evidence from Pakistan and the United States shows that when schools shut down, students often move to lower-performing alternatives, leading to notable declines in learning (Engberg et al. 2012; Andrabi et al. 2020; Larsen 2020). Teachers as well as school owners, who often rely on these schools as their main or only source of income, are affected as well.

In order to address these challenges - financial instability, exposures to shocks, and limited capacity, governments and donors have experimented with a wide range of interventions including financial support and access to ‘Education Products and Services (EPS)’ - such as textbooks, technology, and teacher training. The LEAPS project gave experimental evidence from Pakistan that unconditional grants can boost enrollment, improve infrastructure, and reduce fees among treated schools (Andrabi et al. 2020). However, the focus of this study remained on market equilibrium effects and price and quality responses - not survival. More recent work provides direct evidence on school survival: Andrabi et al. (2025) find that both loans and EPS reduce closure rates by 8–9 percentage points in ITT estimates, with even larger effects among compliers (20–33 percentage points), and that EPS begins to lower closures almost immediately while the impact of loans emerges only after several years. It is important to note, however, that the interventions examined - loans and EPS - differ significantly from unconditional grants in both purpose and mechanism. As a result, the findings do not speak about how grants affect untreated competitors. Moreover, the focus on both of these works was on treated schools themselves, not taking into account the experience of untreated competitors and the local market structure in play. Despite the gaps, these works highlighted the way in which interventions can meaningfully improve school volatility, while reshaping the competitive environment by impacting the fees charged by schools. This is mirrored in broader SME literature, where capital injections help treated firms grow while imposing competitive strains on untreated firms in the same markets. (de Mel, McKenzie & Woodruff 2012; Rotemberg 2014).

Across the literature, it is found that interventions do not land in a vacuum. They are introduced into small, localized markets where schools watch each other closely and react strategically. Drawing on industrial organization models, researchers describe private schools as competing under capacity constraints, differentiated quality, and downward-sloping demand (Kreps & Scheinkman 1983; Bau 2022). When an intervention lowers costs for one school, others respond by lowering fees and marginal costs or by shutting down. This sensitivity to competitive pressure is consistent with evidence on firm survival, highlighting how limited financial buffers can amplify the impact of shocks (McKenzie & Paffhausen 2019).

Yet despite the growth of this literature, two gaps remain. Firstly, the focus of most studies is solely on treated schools: how their enrollment, investments, test scores, or profits change after receiving support. Much less is known about how these programs affect untreated schools sharing the same market, and in particular whether financial interventions raise their risk of closure. Second, while theory suggests that spillover effects should vary with the structure of the local market - for instance, whether the market is highly concentrated or not - little empirical work explicitly connects displacement risks to these indicators of market concentration.

This study speaks directly to both gaps. It shifts attention away from treated schools and toward untreated schools located in treated villages, asking whether unconditional grants affect their likelihood of exit. It also places market structure at the center of the analysis by constructing a Herfindahl–Hirschman Index (HHI) from enrollment shares to capture the intensity of local competition. Drawing reference from a simple game-theoretical model in which schools decide whether to stay or exit after facing asymmetric cost shocks, the paper tests whether untreated schools in more competitive markets are especially vulnerable when their rivals receive grants. In doing so, it links financial interventions, market concentration, and stay-or-exit decisions in a unified framework, and provides new evidence on how policy-induced changes ripple through local schooling markets rather than affecting treated schools alone.

III. Theoretical Framework:

This paper is based in the Industrial Organization theory of firm behavior, under conditions of competition and asymmetric shocks. This framework allows private schools to be conceptualized as strategic agents operating within local markets as ‘small-medium enterprises’. These local markets are characterized as having varying degrees of competition, barriers to entry, as well as financial constraints. When financial incentives, such as grants, are provided to firms - they act as exogenous cost shocks which shift market equilibrium affecting all schools within.

Being portrayed as SMEs, these schools, acting as profit maximizing firms, face downward sloping demand curves, which are determined by tuition, quality of education, and accessibility. They face different degrees of variable as well as fixed cost, which need to be sustainable in the long run for their long-term survival. The profit function of schools can be expressed as:

$$\pi_i = P_i(Q_i) \cdot Q_i - C_i(Q_i, K_i)$$

where P denotes tuition fees, Q denotes enrollment, and C is the cost function dependent on enrollment and investment K . In this example, financial incentives such as grants serve as positive cost shock, shifting the supply curve to the right. In the case where the grant acts as a tool to enable investments and improve quality, the demand curve may shift outwards.

Following up on the experiment carried out in Andrabi et al. (2020), “Upping the Ante: The Equilibrium Effects of Unconditional Grants to Private Schools,” this paper divides our schools into treated and untreated arms - those that receive grants and those that do not. Since untreated schools may find it difficult to compete, especially given their higher marginal costs - they face strategic decisions: whether to stay or exit the market?

These decisions are based on the degree of competition within the market. A highly competitive market that has low concentration, might allow schools to compete using non-price factors. On the other hand, high concentration markets might induce exits among weaker rivals. To calculate

the degree of competition, this paper seeks to utilize the Hirfindahl-Hirschman Index, which sums the squares of market shares. This is shown below:

$$HHI_v = \sum_{i=1}^N s_{iv}^2 \quad \text{where } s_{iv} = \frac{\text{Enrollment}_{iv}}{\sum_{j=1}^N \text{Enrollment}_{vj}}$$

This would help with analysing the decisions made by schools as well as be used as a variable in the regression design. The model would test spillover effects across markets with different baseline HHI levels.

Building on these theoretical insights, the following hypothesis emerges:

H₀: Grant interventions increase the likelihood of untreated schools to exit,

H_{A1}: Grant interventions do not increase the likelihood of untreated schools to exit

H₀: Exit decisions vary according to the village-level market structure

H_{A1}: Exit decisions do not vary according to the village-level market structure

To test these hypotheses, this study will use a panel Difference-in-Differences (DiD) model with school and time fixed effects, allowing for comparison of untreated schools against treated schools. The validity of this design rests on the parallel trends assumption, which requires that in the absence of treatment, treated and untreated schools would have demonstrated comparable trends in the outcome variables.

$$Exit_{it} = \beta_0 + \beta_1 (TreatedVillage_v \times Post_t) + \beta_2 HHI_v + \beta_3 (TreatedVillage_v \times Post_t \times HHI_v) + \mu_i + \tau_t + \epsilon_{it}$$

where $Exit_{it}$ equals 1 if school i in village v exits in period t . $TreatedVillage_v$ indicates whether a village contains a grant-receiving school. $Post_t$ marks periods after the grant, and HHI_v measures baseline market concentration. School fixed effects μ_i control for time-invariant characteristics,

while time fixed effects τ_i capture broader economic or policy changes. The coefficient β_1 measures the average spillover effect of grants on untreated schools, while β_3 captures how this effect varies by market structure. A significant positive β_1 would indicate spillover effects leading to higher exit probability among untreated schools.

IV. Data

This paper draws on data from the Learning and Educational Achievement in Pakistan Schools (LEAPS) project, a multi-round experimental survey conducted in rural Punjab, the largest province in terms of population in Pakistan. The dataset comprises of schools across 266 villages within the district of Faisalabad, Pakistan that had at least two running schools. The data primarily includes 822 private schools as well as 33 NGO schools. The LEAPS project, through a randomised control trial, aimed to evaluate the impact of unconditional financial incentives to these private schools.

The data was collected between July 2012 and November 2014 through a baseline survey, a detailed follow up round after eight months, and subsequent follow-up rounds conducted every three to four months. The panel structure of the data records factors such as operations, finances, level of infrastructure, as well as closure over time. This allows for a holistic view of the impact of introducing financial interventions on over 855 schools through 4275 school-round observations. Each observation corresponds to the survey results from consenting schools in a given round.

The following is a table describing the different variables used in the empirical analysis throughout this paper.

Table 1. Variable Descriptions

Variable	Description
exit	Binary dependent variable equal to 1 if a school faced closure between consecutive survey rounds; 0 otherwise.
post	Indicator equal to 1 for post-grant rounds and 0 for pre-grant rounds.
treat_vil	Village-level treatment intensity: 1 = Control, 2 = Low saturation, 3 = High saturation.

treat_grant	Binary indicator equal to 1 if a school directly received a grant; 0 otherwise.
grant_percap	Grant amount per enrolled student at baseline (PKR).
HHI_{0_loo}	Leave-one-out Herfindahl–Hirschman Index calculated from squared enrollment shares excluding the school’s own share.
schoolage₀	Number of years since school establishment at baseline.
numteachers₀	Number of teachers employed at baseline
enrol₀	Total student enrollment at baseline.
fee₀	Average monthly tuition fee (PKR) at baseline.
invest_index	Mean of seven binary indicators (library, computer, sports facility, boundary wall, fans, electricity, transport) to capture baseline infrastructure quality.

The dependent variable in this study is school *exit*, which has a binary indicator equivalent to 1 if a school faced closure between survey rounds and 0 otherwise. This serves as the central outcome through which the effects of grants are studied. As for the independent variables, the primary variables are the post-grant period (*post*) and treatment exposure (*treat_vil*). This provides insights into post intervention results as well as the intensity of treatment status for each village - control, low saturation, or high saturation.

The binary variable *treat_grant* is used to determine whether a school directly received a grant or not, marking the difference between direct and indirect exposure; whereas, the variable *grant_percap* measures the amount per enrolled student to account for variation in grant intensity. These are valued in the local currency - Pakistani Rupees.

The Herfindahl–Hirschman Index serves as a way to compute the degree of competition within markets. To account for the impact on survival rates for schools, another variable that was created is *HHI_{0_loo}*. This is done by calculating the squared sum of enrollment shares within each village, and excluding the school’s own share to avoid correlation.

On top of this, several baseline school characteristics are included as control variables to capture the heterogeneity in institutional quality and capacity including *schoolage₀*, *numteachers₀*, and

*enrol*₀. Another variable that was constructed is the *invest_index*, which represents the mean of seven binary indicators: presence of a library, computer, sports facility, boundary wall, fans, electricity, and transport access. This summarises the physical quality and the level of infrastructure available at the school.

These summary statistics in the table below provide an overview of the distribution and variation of these variables.

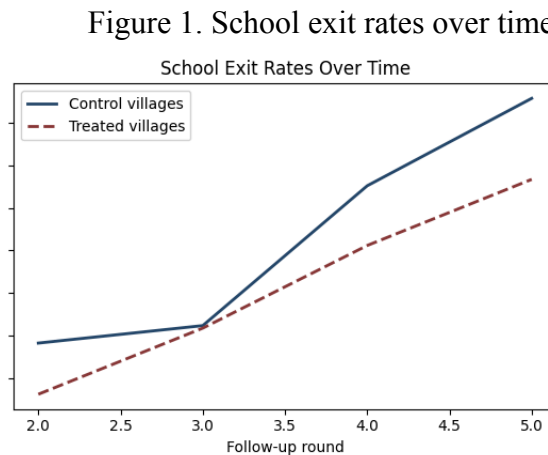
Table 2. Summary Statistics

Variable	Description	Obs	Mean	Std. Dev.
exit	Binary Variable: Indicator = 1 if school closed in a given round	4,275	0.072	0.259
grant_percap	Grant amount per enrolled child (PKR)	4,275	153.70	190.33
treat_grant	Binary Variable: Indicator = 1 if school received a grant	4,275	0.40	0.49
treat_vil	Village treatment status (1=Control, 2=Low, 3=High saturation)	4,275	2.15	0.84
schoolage ₀	Years since school establishment (baseline)	4,260	8.28	6.67
numteachers ₀	Number of teachers at baseline	4,255	8.16	4.85
enrol ₀	Total student enrollment at baseline	4,255	163.64	115.92
fee ₀	Average tuition fee (PKR)	4,255	238.13	165.86
HHI ₀	Market concentration index	4,255	0.067	0.027
invest_index	Infrastructure index (mean of 7 facilities)	4,255	0.567	0.155

The dependent variable *exit* has a mean of 0.072, which reflects that approximately 7% of school-round observations correspond to closures. The treatment variables show that around 40% of schools received a grant, with most of these schools being located in either low-or high-saturation treatment villages. The average grant amount per student is PKR154, and the substantial standard deviation reflects heterogeneity in school size and grant allocation. The average *invest_index* value of 0.57 indicates that most schools do have a majority of the infrastructural facilities that were indexed.

As for the market factors, the mean value of *HHI*₀ (leave-one-out) is 0.067, which suggests moderate market competition; whereas the average fee of PKR 238 reflects that most of the schools are relatively small in size.

Figure 1 below plots school exit rates across survey rounds, separately for treated and control villages. Exit remains relatively low in the initial rounds but rises sharply after round 3 - roughly a year after the baseline survey.



Empirical Analysis

To estimate the impact of unconditional grants on school survival, this paper employs a difference-in-differences (DiD) framework. Three different sets of regressions are run, which account for two varying models: i) an analysis sample consisting of untreated private schools and ii) the full sample of all schools. All of these regressions are estimated using fixed-effects to

absorb time-invariant characteristics of schools as well as to account for common economic and policy shocks across the wave of surveys. Standard errors are clustered at the village level, consistent with village-level treatment assignment.

The first estimates a baseline DiD model, which is reflected below:

$$Exit_{it} = \beta_0 + \beta_1 Post_t + \beta_2 TreatedVillage_i + \beta_3(TreatedVillage_v \times Post_t) + \mu_i + \tau_t + \epsilon_{it}$$

The results from this regression model are summarised below:

Table 3. Regression 1

Variable	Model 1: Untreated Private Schools	Model 2: All Schools	Interpretation
Post (β_1)	0.091*** (0.018)	0.094*** (0.018)	School exits rose by approximately 9 percentage points following the grant period, indicating greater turnover in the schooling market after the intervention.
Post $\times$ Treated Village (β_3)	-0.017 (0.022)	-0.028	Schools in treated villages experienced a 1.7–2.8 percentage-point smaller increase in exit after the grant period.
R ² (within)	0.052	0.048	Around 5% of within-school variation in exit is explained by the model.
ρ (rho)	0.77	0.75	Approximately 75% of variation is between schools

Notes: Robust standard errors clustered at the village level. * $p < 0.01$, $p < 0.05$, $p < 0.10$.

Dependent variable = 1 if a school closed in a given round.

The term TreatedVillage was dropped since it was collinear with fixed effects.

In both specifications, the post coefficient (β_1) is positive and highly significant, showing that school closures increased substantially after the grant cycle, regardless of treatment status. For both, the analysis sample of untreated private schools and the full sample, the results are similar. Although the unconditional mean exit rate in the sample is 7%, the coefficient on *post* suggests that exit rates increased by 9% in the post-treatment period.

The interaction term is negative, which implies that schools located in treated villages were less likely to close compared to those in control villages, albeit the difference is marginal.

While the baseline model provides a useful benchmark, it does not account for underlying heterogeneity across schools. The second specification therefore adds a set of baseline controls to reflect how exit rates vary. The results for these are reflected below:

Table 4: Regression 2

Variable	Model 1: Untreated Private Schools	Model 2: All Schools	Interpretation
Post (β_1)	0.127*** (0.032)	0.107*** (0.031)	School exits increased by roughly 10–13 percentage points after the grant cycle, indicating a more volatile post-intervention period.
Post $\times$ Treated Village (β_3)	−0.018	−0.016 to −0.029	Exit rates rose slightly less in treated villages
Post $\times$ (Invest_index)	0.059	−0.086** (0.04)	For all schools, better baseline infrastructure reduced closure risk post-grant
Post $\times$ (Enrol ₀)	0.00012	0.000095	Baseline enrollment levels do not significantly predict closure once fixed effects are controlled for.
Post $\times$ (Schoolage ₀)	0.0034*** (0.001)	−0.0020** (0.016)	For untreated schools, older institutions were more likely to close; for all schools, older ones were more stable.
R ² (within)	0.063	0.057	Consistent with baseline models; around 6% of within-school variation explained.
ρ (rho)	0.77	0.74	Roughly 75% of exit variance occurs between schools, consistent with persistent structural differences.

Notes: Robust standard errors clustered at the village level. * $p < 0.01$, $p < 0.05$, $p < 0.10$.

Dependent variable = 1 if a school closed in a given round.

The results of this model are consistent with the earlier baseline model, yet they reveal the ways in which heterogeneity amongst schools affects outcomes. The coefficient for *post* is positive and statistically significant again, yet the magnitude has risen compared to our baseline model - between 10.7 to 12.7 percentage points.

As for the interaction term between *post* and *invest_index*, the coefficient is negative and statistically significant in the full sample. This suggests that the risk of closure was lower for schools with stronger baseline infrastructure facilities, allowing them to be more resilient. On the other hand, the coefficient was not statistically significant for the analysis sample. This implies that infrastructure differences mattered less for untreated private schools.

The $Post \times schoolage_0$ variable reveals an interesting divergence: older schools in the analysis sample were more likely to close in the post period; whereas, in the full sample, they were less likely to close. Among untreated private schools, these older institutions may have been displaced by newer, better-funded competitors.

The third specification extends the analysis by adding the control for village-level HHI - this is reflected in the regression model below:

Table 5: Regression 3

Variable	Model 1: Untreated Private Schools	Model 2: All Schools	Interpretation
Post (β_1)	0.168 *** (0.005))	0.146 ** (0.013)	School exits increased by roughly 15–17 percentage points after the grant cycle, indicating a more volatile post-intervention period.
Post $\times$ Treated Village (β_2)	−0.014	−0.075 / −0.036	Schools in treated villages faced smaller increases in exit rates
Post $\times$ HHI₀ (β_3)	−0.49	−0.52	More concentrated markets (higher HHI) appear slightly less prone to exit post-grant, but not significantly.
Post $\times$ Treated Village $\times$ HHI₀ (β_4)	−0.13	.86 / 0.10	No significant heterogeneity by market concentration

Post × (Invest_index)	0.05	−0.078** (0.05)	For all schools, better baseline infrastructure reduced closure risk post-grant
Post × (Schoolage₀)	0.003*** (0.001)	−0.002** (0.016)	For untreated schools, older institutions were more likely to close; for all schools, older ones were more stable.
R² (within)	0.065	0.058	Similar fit to prior models; around 6% of within-school variation explained.
ρ (rho)	0.77	0.75	Roughly 75% of exit variance occurs between schools, consistent with persistent structural differences.

Notes: Robust standard errors clustered at the village level. * $p < 0.01$, $p < 0.05$, $p < 0.10$.
Dependent variable = 1 if school closed in a given round.

This specification of the regression model added one more control variable - the village-level Herfindahl–Hirschman Index (HHI). The coefficient on *post* is positive once again and statistically significant across both samples, ranging from 0.146 to 0.168, suggesting that the post-treatment period was associated with a 15–17 percentage-point increase in school closures.

The key heterogeneity term within this model is the triple interaction term $\beta_4(Post \times TreatedVillage \times HHI)$, which turned out to be statistically insignificant in both models. This suggests that the effect of the grants did not vary systematically between more concentrated and more competitive schooling markets.

In this model as well, the coefficient for school age is statistically significant. For the analysis sample, older private schools were once again more likely to shut down, being potentially displaced by newer entrants from the grant environment.

Robustness Checks

The robustness checks were carried out in different steps. First, the Hausman test was used to verify the choice of fixed effects over random effects. Secondly, the Pesaran CD test was used to assess cross-sectional dependence in the panel. Third, the Modified Wald test was used to evaluate heteroskedasticity within the residuals to support the usage of cluster-robust standard

errors. Finally, the variables from the panel data were used on a pooled OLS model to test for multicollinearity.

To compare the consistency of fixed-effects (FE) and random-effects (RE) estimators, a Hausman specification test was conducted on a simplified model, which included only time-varying regressors. The test produced a degenerate result ($\chi^2(6) = 0.00$, $p = 1.00$).

Since this test was inconclusive, additional diagnostics were used, which supported the use of fixed effects. The intra-class correlation ($\rho \approx 0.75$) suggested that approximately three-quarters of the total variation in school exit occurs between schools, which reflects substantial heterogeneity across institutions. Moreover, the F-test for individual effects strongly rejected the null hypothesis that all school-level effects are zero ($F(854,3414) = 14.88$, $p < 0.001$). As a result, the FE model was retained to isolate within-school changes while controlling for time-invariant traits.

Given that schools within the same local markets may be exposed to economic and policy shocks, it is important to assess whether residuals in the panel regression are cross-sectionally independent. To test this assumption, the Pesaran (2004) test for cross-sectional dependence was applied to the residuals from the fixed-effects specification. The test strongly rejected the null hypothesis of cross-sectional independence ($CD = 993.59$, $p < 0.001$), with an average absolute residual correlation of $|\bar{\rho}| = 0.95$. This result indicates that the residuals are highly correlated across schools, suggesting the presence of shocks or unobserved market-wide factors that influence school closure decisions across villages.

Heteroskedasticity occurs when the error term does not have constant variance, violating the classical assumption of homoskedasticity and potentially leading to inefficient and biased standard errors. To test for this variance across panel units, a Modified Wald test for groupwise heteroskedasticity (`xttest3`) was conducted following the fixed-effects regression. This resulted in a high value for $\chi^2(849) = 1,486,989.67$ and a p-value < 0.001 . As a result, the null hypothesis was rejected, which indicated a strong presence of heteroskedasticity.

Given this result, the regressions were estimated using cluster-robust standard errors at the village level. This was done to ensure that the coefficient estimates remain consistent and the reported significance levels reliable, even in the presence of heteroskedasticity and correlated shocks within villages. The use of clustered standard errors thereby preserves the robustness of the Difference-in-Differences estimates.

Finally, to ensure that multicollinearity was not biasing the estimates in the panel fixed-effects regressions, a Variance Inflation Factor (VIF) test was conducted using a pooled OLS specification of the main model variables. However, it must be noted that the tests on the pooled OLS do not reflect the Fixed Effects transformation. The results are shown in the table below:

Table 6. Multicollinearity

Variable	VIF	1/VIF	Interpretation
numteachers ₀	4.08	0.245396	Moderate correlation with other variables
enrol ₀	3.63	0.275193	Moderate correlation with other variables
grant_percap	1.99	0.501334	Low correlation - no concern
treat_grant	1.81	0.552208	Low correlation - no concern
invest_index	1.32	0.758411	Low correlation - no concern
fee ₀	1.31	0.763582	Low correlation - no concern
schoolage ₀	1.10	0.906594	Close to uniform correlation
HHI _{0_loo}	1.09	0.914167	Close to uniform correlation
treat_vil	1.06	0.945809	Close to uniform correlation
Mean VIF	1.84	—	Below conventional thresholds

As shown in the table above, the VIF values for the explanatory variables recorded is well below conventional thresholds. The mean VIF is 1.84. The two highest values are the for the variable *numteachers₀* and the variable *enrol₀*. All other variables had VIFs near unity, which indicates very low correlation among covariates.

V. Conclusion

This paper examined how unconditional grants to private schools reshape survival patterns within local schooling markets. In the post-grant period, closure rates rose sharply regardless of the treatment measure for schools. Similarly, the degree of market concentration did not play a meaningful role in moderating these effects; rather, the forces at play were fundamentally more school-specific. The heterogeneity patterns revealed how baseline infrastructure emerged as a key determinant of resilience - with better resourced schools having a higher likelihood of survival. On the other hand, long-standing institutions were more likely to shut down and fail to adapt when faced with new entrants who received financial support.

This study, thereby, provides meaningful insights into market dynamics. For those schools that already have a strong foundation, grants may allow them to leverage improvements. At the same time, institutions with structural weaknesses, such as outdated infrastructure or high fixed costs accumulated over many years of operation, were more vulnerable to exit. This shows that interventions alone do not ignite or trigger displacements through market structures, but through the ability of schools to differentiate themselves and absorb competitive pressure.

The results shed light on the need for a wider evaluation of incentive tools available to policymakers. While the main focus of this paper was unconditional grants, the patterns observed may vary depending on the type of measure employed. Grants reduce immediate cost pressures and help schools upgrade quality, but they may magnify pre-existing disparities across institutions. Other instruments, such as loans or performance payments, might operate through different channels: liquidity support, credit constraints, or demand-side responses. Grants did not, on average, displace untreated schools. Yet, the fact that they altered the distribution of survival across school types suggests that other interventions could have analogous effects, depending on how they ease constraints or influence demand. It then becomes important to understand how each instrument interacts with underlying school vulnerabilities.

Nevertheless, the evidence presented here should be interpreted with caution. Although the dataset follows schools across five survey rounds, the two-year post-intervention window remains short relative to how survival dynamics typically unfold in this sector. Low-cost private schools operate in highly volatile environments, where closures and recoveries often respond to

longer-term shifts in demand, demographics, and external shocks. Due to the short term of these survey results, the analysis only captures the immediate adjustment period, not the long-run equilibrium. Moreover, certain baseline characteristics may also proxy underlying dimensions of quality that cannot be observed or quantified directly.

In conclusion, this paper helped identify patterns that underscore the complexity of designing policies in fragmented and highly volatile education markets, where even well-intentioned support can magnify existing disparities. The findings offer a foundation for further research that tracks schools over longer periods and compares a wider range of incentive tools to deepen the understanding of what effective support can look like that ripples through schooling ecosystems.

Appendix 1. A Game Theory Model

This appendix presents a simple model of school competition and exit in the presence of asymmetric financial support. This aims to extend the theory and clarify the mechanisms through which grants to some schools can affect the survival of others. Instead of trying to fully characterize and evaluate all possible equilibria, this model provides the motivation behind the empirical specification.

Consider a local market with two private schools, indexed by $i = 1, 2$, that operate under capacity and credit constraints. School i is characterised by a quality level $q_i \in (0,1)$ and a fixed capacity $x_i \in (0,1)$; we denote its type by $i = (q_i, x_i)$. Each school incurs a fixed operating cost $F_i(q_i, x_i)$, which captures rent, utilities, and other overhead costs. For the sake of simplicity, marginal costs are normalised to zero. Capacity and quality are predetermined and cannot be freely adjusted because of financial constraints.

The time period is split into three periods. In period 1, an external actor allocates financial support: one school may receive a grant, while the other remains untreated. In period 2, a negative demand shock may occur, and schools must decide whether to remain open or exit the market. If they remain open, schools simultaneously choose tuition fees. Finally, in period 3, surviving schools earn continuation payoffs but make no further decisions.

There is a unit mass of students indexed by j . Each student j demands one seat and has a preference for quality, θ_j , drawn uniformly from $[0,1]$. A student who attends school i and pays fee p_i obtains utility:

$$U_j(\theta_j, q_i, p_i) = U_j q_i - p_i$$

and the alternative of not attending any school gives them zero utility.

Based on economic theory, traditional models yield a demand system in which each school faces a downward-sloping residual demand curve. When both schools are active, school i chooses its fee to maximise one-period profit. Let

$$p_i(q_i, x_i; q_{-i}, x_{-i})$$

denote the resulting equilibrium fee of school i in the absence of aggregate shocks. Given this fee, school i enrolls x_i students and earns profit from period 2 given by:

$$\pi_i^0(q_i, x_i) = x_i p_i(q_i, x_i; q_{-i}, x_{-i}) - F_i(q_i, x_i)$$

In period 1, the treated school receives a grant of size $K > 0$, which is modelled as a reduction in fixed cost:

$$F_i^T(q_i, x_i) = F_i(q_i, x_i) - K$$

while the untreated school faces

$$F_i^U(q_i, x_i) = F_i(q_i, x_i)$$

In period 2, the village experiences a negative demand shock with probability $\mu \in (0,1)$. The shock reduces willingness to pay for private schooling by a scalar $\varepsilon > 0$. Under this shock, the equilibrium fee of school i shifts down to

$$p_i^\varepsilon(q_i, x_i) = p_i(q_i, x_i; q_{-i}, x_{-i}) - \Delta_i(q_i, x_i, \varepsilon)$$

With $\Delta_i > 0$, and profits become

$$\pi_i^\varepsilon(q_i, x_i) = x_i p_i^\varepsilon(q_i, x_i) - F_i^\varepsilon(q_i, x_i)$$

where $g \in \{T, U\}$ denotes whether the school is treated or untreated. We assume that in the absence of the demand shock, both schools would strictly prefer to continue operating:

$$\pi_i^0(q_i, x_i) > \pi^0$$

where $\pi^0 \geq 0$ is the minimum payoff the owner requires to justify keeping the school open.

In period 2, after observing whether the demand shock occurs, each school faces a dilemma and chooses whether to stay or exit. Let $d_i \in \{0, 1\}$ denote the closure decision of school i , where $d_i = 1$ corresponds to exit. A school remains open in period 2 if and only if its profit in that period weakly exceeds the outside option:

$$\pi_i^\varepsilon(q_i, x_i) \geq \pi^0$$

and exits otherwise. This condition defines a “survival region” in the (q_i, x_i) space.

The first observation is that, holding the competitive environment fixed, vulnerability to exit is systematically related to quality and capacity.

Proposition 1. Suppose period-2 profits $\pi_i^\varepsilon(q_i, x_i)$ are strictly increasing in both quality q_i and capacity x_i . Then there exists a frontier in (q, x) - space such that, for a given treatment status and demand realisation, schools with sufficiently low quality and capacity lie below the frontier and exit, while schools with higher quality or greater capacity survive. In particular, if a school of type (q, x) exits in equilibrium, then any school facing the same environment with weakly lower quality or weakly lower capacity also chooses to exit.

This result is based on the idea that schools that are smaller or weaker along key baseline dimensions are more likely to close when hit by adverse shocks, conditional on the market

environment. In the analysis above, this maps into systematic differences in exit risk by baseline characteristics such as infrastructure, enrolment, or age.

Another observation looks into the direct effect of financial support on these survival thresholds. Consider two otherwise identical schools facing the same demand shock, one treated with a grant and one untreated.

Proposition 2. For any given (q, x) and demand, the difference in period-2 profits between a treated and an untreated school of type (q, x) equals the grant amount K . Consequently, whenever a treated school of type (q, x) exits, it is also optimal for an untreated school of type (q, x) to exit. The inverse of this relationship need not hold.

The grant, therefore, expands the set of states under which a school survives: for schools close to the exit threshold, the difference between continuation and closure can lie on financial support.

Finally, it is important to factor in the market structure to these individual decisions. Let market structure be summarised by a concentration parameter H , interpreted as a Herfindahl–Hirschman–type index, whereby lower values correspond to more competitive markets. A higher degree of competition (lower H), thus, tends to reduce markups and profits for all schools. When one school receives a grant, this effect interacts with the treatment status of its rival.

Proposition 3. Suppose that (i) equilibrium profits of untreated schools are decreasing in the degree of competition (lower H implies lower π_i^ε), and (ii) the grant to the treated school leads to a reduction in its equilibrium fee or an improvement in perceived quality. Then the survival threshold for untreated schools depends on both treatment exposure and market structure. With lower degree of competition, an untreated school may choose to stay; however with a strong degree of market concentration and facing ‘treated’ rivals, the school may inevitably choose to exit.

This proposition shows that asymmetric support interacts naturally with local market structure, creating implications for how probabilities for these stay-or-exit decisions may vary according to the degree of competition faced.

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